ECON-7102: Environmental Economics I

Class 14: Property value models and Chay & Greenstone (2005)

Casey J. Wichman

School of Economics
Georgia Institute of Technology

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Today's agenda

1. Questions?
2. Housekeeping:
   - Abstract 3: rapid-fire pitches today.
   - Midterm: Wed Oct 21, in class.
3. Today:
   - The hedonic model overview
   - Chay and Greenstone (JPE, 2005)
4. For next class...

A puzzle: does air quality matter for house prices?

- Regress log county house values on TSPs (total suspended particulates).
- Cross-section and first-difference estimates are small, unstable across specifications, and often *positive*.
  - Smith and Huang (1995): about a quarter of published hedonic air quality estimates have the "perverse" sign.
- **Q:** Are people WTP more for pollution? What might be biasing these estimates?

TABLE 3
Cross-Sectional and First-Difference Estimates of the Effect of TSPs Pollution on Log Housing Values

	(1)	(2)	(3)	(4)
	A. 1970 Cross Section			
Mean TSPs (1/100)	.032	−.062	−.040	−.024
	(.038)	(.018)	(.017)	(.017)
R^2	.00	.79	.84	.85
Sample size	988	987	987	987
	B. 1980 Cross Section			
Mean TSPs (1/100)	.093	.096	.076	.027
	(.066)	(.031)	(.030)	(.028)
R^2	.00	.82	.89	.89
Sample size	988	984	984	984
	C. 1970–80 (First Differences)			
Mean TSPs (1/100)	.102	.024	.004	−.006
	(.032)	(.020)	(.016)	(.014)
R^2	.02	.55	.65	.73
Sample size	988	983	983	983
County Data Book covariates	no	yes	yes	yes
Flexible form of county covariates	no	no	yes	yes
Region fixed effects	no	no	no	yes

Note.—See the notes to tables 1 and 2. For 1970 and 1980, the mean TSPs variable is the 1969–72 and 1977–80 average of the annual geometric mean concentrations, respectively. See the Data Appendix for a full list of the control variables. The flexible functional form includes quadratics, cubics, and interactions of the variables as controls. The mean of the natural log of 1970 housing prices is 10.55. The means of the dependent variables in panels B and C are 10.82 and 0.27, respectively. Standard errors (in parentheses) are estimated using the Eicker-White formula to correct for heteroskedasticity.

The hedonic model in a nutshell

- A house is a bundle of characteristics x and environmental quality q. Its market price is the **hedonic price function** $P(x, q)$.
- Households choose a house, and therefore choose q:

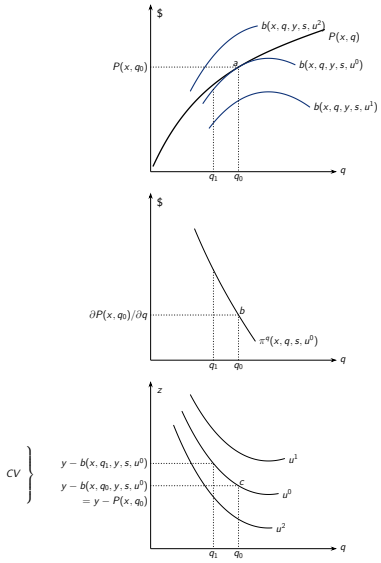
$$\max_{x,q,z} U(x, q, z; s) \quad \text{s.t.} \quad y = P(x, q) + z$$

- FOCs for q and z give

$$\text{MWTP for } q \equiv \frac{\partial U/\partial q}{\partial U/\partial z} = \frac{\partial P(x, q)}{\partial q} \equiv \text{implicit price of } q$$

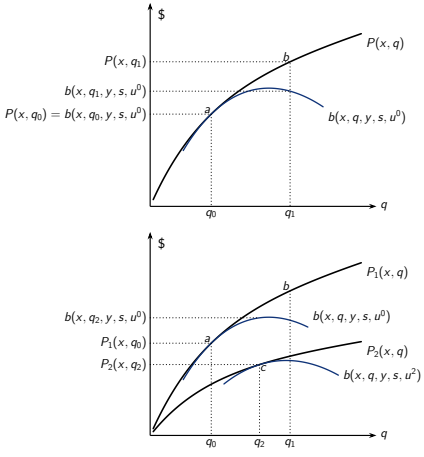
$\Longrightarrow$ **At the house a household chose, the slope of $P(x, q)$ equals that household's MWTP for q.**

- Top panel: the bid function $b(\cdot)$ is an indifference curve in $(q, \$)$ space, tangent to $P(x, q)$ at the chosen point a.



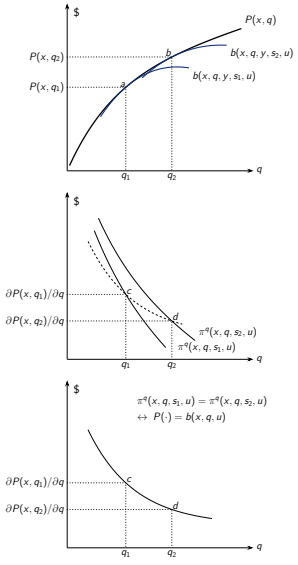
From the hedonic price gradient to welfare

- **Marginal or localized change in q** (top):
  - $P(x, q)$ does not shift. Occupant's own WTP if they stay:
    $b(x, q_1, y, s, u^0) - P(x, q_0) < P(x, q_1) - P(x, q_0)$.
  - Free mobility: the occupant moves to another q_0 house (back to *a*), and the owner gains $P(x, q_1) - P(x, q_0)$. No one's utility changes, so the price change is the welfare gain.
  - With large moving costs: the occupant stays at *b*. The net gain is $b(x, q_1, y, s, u^0) - P(x, q_0)$, which is less than $P(x, q_1) - P(x, q_0)$.
- **Large change affecting many households** (bottom):
  - $P(x, q)$ shifts from P_1 to P_2, and households re-sort (point *c*). Welfare needs the MWTP *function*, not one point on it.
  - *Intuition:* a city-wide cleanup makes clean air less scarce, so its implicit price falls. A house's price change no longer measures WTP. Part of it is a transfer between buyers and sellers.



What does a cross-section of house prices identify?

- Each observed sale (points *a*, *b*) is a tangency for a *different* household.
  - $P(x, q)$ is the envelope of many bid functions. It is not anyone's demand curve.
  - Points *c* and *d* lie on different households' MWTP functions (panel b).
- **Problem 1: recovering the MWTP function (2nd stage)**
  - Can make (strong) assumptions about prefernces to recover the full MWTP function.
  - Assume identical preferences (panel c) or multiple markets are the usual fixes.
- **Problem 2: estimating the slope of $P(x, q)$ (1st stage)**
  - Polluted places are also more urban, denser, higher income, and higher crime.
  - Omitted variables bias $\partial P/\partial q$ directly, which could explain the cross-sectional puzzle.
- Problem 2 is the motivation for Chay and Greenstone (2005).



Chay & Greenstone (2005): setting and data

Goal: estimate the hedonic gradient ($\partial P/\partial q$) for air quality, and the average MWTP allowing for sorting.

- **Clean Air Act Amendments of 1970** set national standards (NAAQS) for TSPs. A county is **nonattainment** if either:
  - annual geometric mean TSPs $> 75\ \mu g/m^3$, or
  - second-highest daily TSPs $> 260\ \mu g/m^3$.
- States must impose stricter controls on polluters in nonattainment counties. EPA can enforce with penalties.
- **Data:** county level, 1970 and 1980.
  - TSPs from every EPA monitor (obtained by FOIA request)
  - Median house values and county characteristics from the Census and County Data Books
  - About 988 counties with monitor coverage

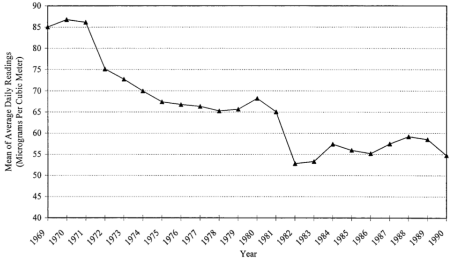


FIG. 1.—National trends in TSPs pollution, 1969–90. The data points are derived from the 169 counties that are continuously monitored in this period. These counties had a total population of approximately 84.4 million in 1980. The annual county means were calculated as the weighted average of the monitor-specific geometric means, where the weight is the number of monitor observations. The year-specific average is calculated as the weighted average of the county-specific means, where the weight is the 1980 population.

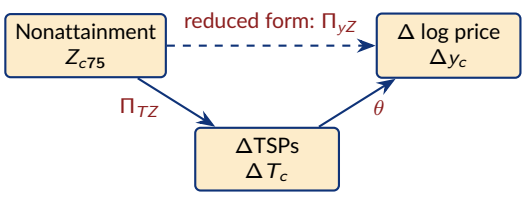
Empirical strategy

- **Cross-section:** $y_{c70} = X'_{c70}\beta + \theta T_{c70} + \varepsilon_{c70}$, with $\varepsilon_{c70} = \alpha_c + u_{c70}$
- **First differences** absorb permanent county effects α_c:

$$\Delta y_c = \Delta X'_c \beta + \theta \Delta T_c + \Delta u_c$$

  - Still biased if shocks to pollution covary with shocks to house prices (e.g., local economic growth).
- **IV:** instrument ΔT_c with $Z_{c75} = 1(\text{nonattainment in 1975}) = 1(T_{c74} > \bar{T})$

- First stage: $\Delta T_c = \Delta X'_c \Pi_{TX} + Z_{c75}\Pi_{TZ} + \Delta v_c$
- Reduced form: $\Delta y_c = \Delta X'_c \Pi_{yX} + Z_{c75}\Pi_{yZ} + \Delta u_c$
- $\hat{\theta}_{IV} = \hat{\Pi}_{yZ} / \hat{\Pi}_{TZ}$
- Exclusion: $E[Z_{c75}\, \Delta u_c] = 0$
- Robustness: regression discontinuity near 75 μg/m^3



First stage and reduced form, by 1974 TSPs

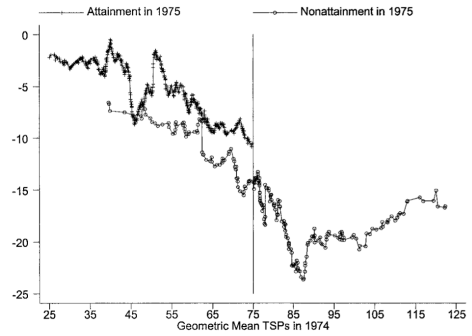


FIG. 4.—1970–80 change in mean TSPs by 1975 nonattainment status and the geometric mean of TSPs in 1974.

- Nonattainment counties had larger 1970–80 declines in TSPs.

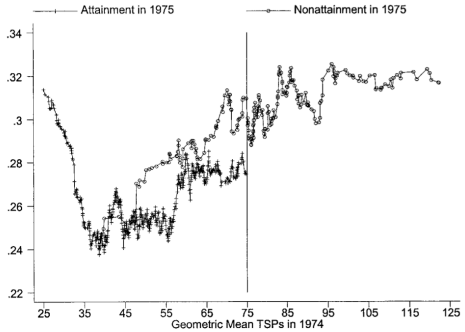


FIG. 5.—1970–80 change in log housing values by 1975 nonattainment status and the geometric mean of TSPs in 1974.

- The same counties had larger 1970–80 increases in log house values.

Results: reduced form

- Nonattainment reduced mean TSPs by about 9–10 $\mu g/m^3$ relative to attainment counties (panel A).
- Nonattainment raised log house values by about 0.02–0.036, or 2–3.6% (panel B).
- Both are policy-relevant on their own: the effect of the regulation on air quality and on property values.

TABLE 4
ESTIMATES OF THE IMPACT OF MID-DECADE TSPs NONATTAINMENT ON 1970–80 CHANGES IN TSPs POLLUTION AND LOG HOUSING VALUES

	(1)	(2)	(3)	(4)
	A. Mean TSPs Changes			
TSPs nonattainment in 1975 or 1976	−9.96	−10.41	−9.57	−9.40
	(1.78)	(1.90)	(1.94)	(2.02)
F-statistic TSPs nonattainment*	31.3	29.9	24.4	21.5
	(1)	(1)	(1)	(1)
R^2	.04	.10	.19	.20
	B. Log Housing Changes			
TSPs nonattainment in 1975 or 1976	.036	.022	.026	.019
	(.012)	(.009)	(.008)	(.008)
F-statistic TSPs nonattainment*	8.5	6.2	9.3	6.4
	(1)	(1)	(1)	(1)
R^2	.01	.56	.66	.73
County Data Book covariates	no	yes	yes	yes
Flexible form of county covariates	no	no	yes	yes
Region fixed effects	no	no	no	yes
Sample size	988	983	983	983

NOTE.—See the notes to previous tables. In panel A the dependent variable is the difference between the 1977–80 and 1969–72 averages of mean TSPs concentrations. The mean is −7.82 $\mu g/m^3$. In panel B the dependent variable is the difference between 1980 and 1970 log housing values, and its mean is 0.27. Standard errors (in parentheses) are estimated using the Eicker-White formula to correct for heteroskedasticity.

* Numbers in parentheses in rows with *F*-statistics are numerator degrees of freedom.

Results: IV estimates of the hedonic gradient

- A 1 μg/m^3 reduction in TSPs raises house values by 0.2–0.36% (panel A).
  - Elasticity of house values with respect to TSPs: -0.20 to -0.35.
- 5–8 times the 1970 cross-sectional estimates with controls (Table 3), and much less sensitive to specification.
- Panel C instruments with 1970–72 nonattainment. Those counties also had weaker 1970s economies, so C&G expect a positive bias. Estimates are small or wrong-signed.
- **Q:** Which point on the hedonic figure does $\hat{\theta}_{IV}$ correspond to? Whose MWTP is it?

TABLE 5
INSTRUMENTAL VARIABLES ESTIMATES OF THE EFFECT OF 1970–80 CHANGES IN TSPs POLLUTION ON CHANGES IN LOG HOUSING VALUES

	(1)	(2)	(3)	(4)
	A. TSPs Nonattainment in 1975 or 1976			
Mean TSPs (1/100)	−.362	−.213	−.266	−.202
	(.152)	(.096)	(.104)	(.090)
Sample size	988	983	983	983
	B. TSPs Nonattainment in 1975			
Mean TSPs (1/100)	−.350	−.204	−.228	−.129
	(.150)	(.099)	(.102)	(.084)
Sample size	975	968	968	968
	C. TSPs Nonattainment in 1970, 1971, or 1972			
Mean TSPs (1/100)	.072	−.032	−.050	−.073
	(.058)	(.042)	(.041)	(.035)
Sample size	988	983	983	983
County Data Book covariates	no	yes	yes	yes
Flexible form of county covariates	no	no	yes	yes
Region fixed effects	no	no	no	yes

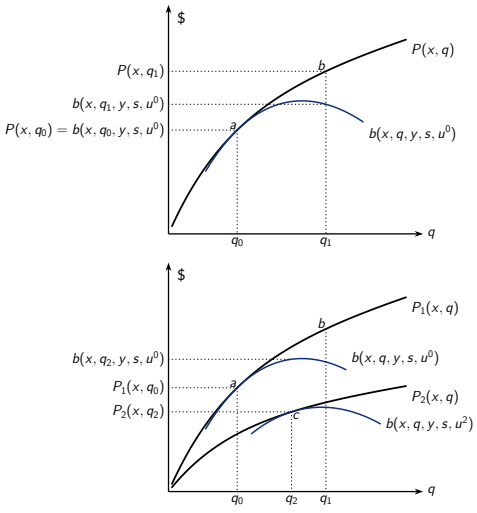
NOTE.—See the notes to previous tables. The coefficients are estimated using 2SLS. The first row of panels A–C indicates which instrument is used. From panels A to C, the instruments are an indicator equal to one if the county was nonattainment for TSPs in either 1975 or 1976, an indicator equal to one if the county was nonattainment for TSPs in 1975, and an indicator that equals one if the county was nonattainment for TSPs in either 1970, 1971, or 1972, respectively. Standard errors (in parentheses) are estimated using the Eicker-White formula to correct for heteroskedasticity.

Heterogeneous preferences and sorting

- If tastes for clean air are identical, the gradient θ equals average MWTP for everyone.
- With heterogeneous tastes: $\Delta y_i = \theta_i \Delta T_i + \varepsilon_i$, with $E[\theta_i] = \bar{\theta}$.
  - Least squares is biased if $E[\varepsilon_i T_i] \neq 0$ (omitted variables) or if θ_i covaries with T_i (self-selection).
  - If households with a weak distaste for pollution sort into dirty counties, MWTP is lower where TSPs are high.
- C&G estimate a random coefficients model and find modest evidence of this pattern.
  - Implication: the benefit of a TSP reduction may be *larger* in cleaner counties (since nonattainment status target the most polluted counties).
- **Other readings for today pick up from here:**
  - Banzhaf and Walsh (AER, 2008): do households move in response to changes in local environmental quality? A direct test of Tiebout sorting.
  - Christensen and Timmins (JPE, 2022): housing discrimination constrains where minority households can sort, which shapes pollution exposure.

From capitalization to welfare

- The IV gradient gives average MWTP for a *marginal* change (e.g., a straight-line tangency from q_0 to q_1 in top panel), but the regulation produced a large change.
- C&G assume constant MWTP (homogeneous, linear preferences):
  - About a 10-unit decline in TSPs, capitalized at 0.28% per unit
  - Mean 1970 house value of $86,900 (2001 dollars) $\Rightarrow$ about $2,400 per house
  - About 19 million houses in nonattainment counties
  - $\Rightarrow$ WTP for the 1970s TSP reductions of about **$45 billion**
- C&G (fn. 41) note two biases in opposite directions:
  - Uncompensated MWTP overstates the gain relative to a compensated measure.
  - Ignoring re-sorting and supply responses understates the gain.



Takeaways

- Insight from hedonic model: the slope of $P(x, q)$ is MWTP for q at the chosen point.
- Pollution is correlated with unobserved local economic conditions, so empirically estimating $\partial P/\partial q$ is difficult.
- C&G use a regulatory threshold as a quasi-experiment. Capitalization estimates are much larger and more stable than cross-sectional estimates.
- Moving from capitalization to welfare still requires theory: marginal vs. nonmarginal changes, heterogeneous preferences, and sorting.
- This design (regulatory or natural experiment $\rightarrow$ house prices) has been used many times since.

For next class...

- **Mon Oct 19 (Class 15):** Value of a statistical life.
- **Wed Oct 21 (Class 16):** Midterm, in class.
- Readings and deadlines: course website.

Appendix: bid functions and compensated MWTP

- Let $\bar{u}$ represent a reference level of utility, such that $\bar{u} = U(x, q, z; s)$
  - Also, let $z = U^{-1}(x, q, \bar{u}; s)$

- Define a dollar amount b and attribute levels x and q so that $U(x, q, y - b; s) = \bar{u}$
  - This implicitly defines a bid function $b(x, q, y, s, \bar{u})$ that measures the maximum amount a consumer would pay for a commodity unit with attribute levels x and q holding utility fixed at $\bar{u}$.

    $\implies b(x, q, y, s, \bar{u}) \equiv y - U^{-1}(x, q, \bar{u}; s)$
  - Implicitly differentiating U w.r.t. q provides:

$$0 = \frac{\partial U(\cdot)}{\partial q} - \frac{\partial U(\cdot)}{\partial z}\frac{\partial b(\cdot)}{\partial q} \implies \pi^q(x, q, s, \bar{u}) = \frac{\partial b(x, q, y, s, \bar{u})}{\partial q} = \frac{\partial U(\cdot)/\partial q}{\partial U(\cdot)/\partial z}$$

  - Where $\pi^q(\cdot)$ is the *compensated* MWTP function for q (i.e., holds utility constant)

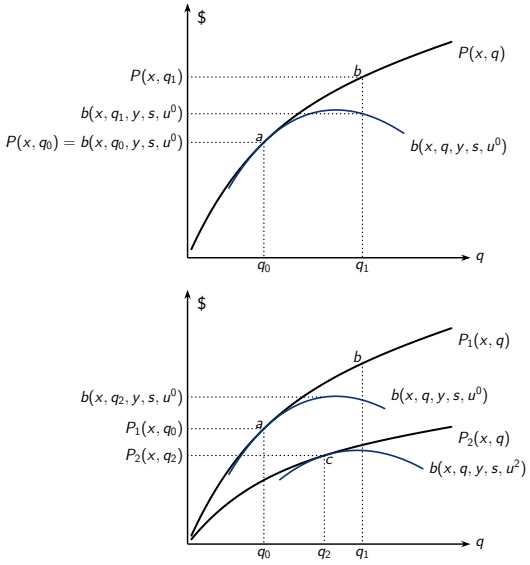
- At the chosen point, $\frac{\partial b(x,q,y,s,u^0)}{\partial q} = \frac{\partial P(\cdot)}{\partial q}$.
- CV for a change from q_0 to q_1: $b(x, q_0, y, s, u^0) - b(x, q_1, y, s, u^0)$, the area under π^q from q_0 to q_1.

Appendix: general equilibrium welfare

- GE allows $P(x, q)$ to shift
- If quality change affects many people, this causes a shift in hedonic price function to $P_2(x, q)$
- New equilibrium (point c) emerges.
- WTP for improvement is given by area under compensated demand function + difference in hedonic price functions:

$$CV^{GE} = \int_{q_0}^{q_2} \pi^q(x, q, s, u^0)\, dq + [P_1(x, q_0) - P_2(x, q_2)]$$

  - Virtually impossible to measure in practice. So, PE framework is often thought of as a lower bound.
  - Moving to new location must be welfare improving.



Appendix: structural approaches

- **Discrete choice models:** Start with specifying a utility function for households, then use utility maximizing behavior to generate estimating equations.
  - Households choose properties among a discrete number of alternatives.
  - Typically estimated using discrete choice, random utility models (RUM)
  - We observe a household's choice among alternatives (researcher specifies choice set)
  - Estimation recovers "structural parameters" of the utility function, which allows us to calculate WTP for changes in q.

- **Sorting models:** Starts with a discrete-choice model, but adds in spatial/housing market equilibrium.
  - Households choose a neighborhood that provides highest utility (e.g., because of green space or school quality)
  - Then, prices adjust until housing market clears.
  - A household's preferred combination of housing/neighborhood characteristics is an equilibrium outcome, conditional on how all other households' decisions affect equilibrium bundle of housing price/characteristics.
  - Relies on assumptions of perfect information / free mobility.

Appendix: quasi-experimental hedonics

- Uses natural experiments to exploit exogenous changes in q.
  - Differences-in-differences
  - Repeat sales (same house, different q)
  - Regression discontinuity (e.g., Chay & Greenstone (JPE, 2005), Black (QJE, 1999) on school quality)
- Largely stems from concerns about omitted-variable bias in (cross-sectional) hedonic estimation.
- Requires fewer assumptions about household preferences
  - Still requires theory to interpret treatment effects as MWTP for changes in q